THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

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LAB 05: CREATE STAR TOPOLOGY (CONFIGURE IP ADDRESSES AND PERFORM PING TEST)

1. INTRODUCTION

This lab implements a star topology LAN in Cisco Packet Tracer with six PCs connected to a central switch. The primary goals are:

- Assign static IP addresses to all PCs.
- Verify connectivity using `ping`.
- Run the simulation to observe and analyze the packet flow from source to destination through the switch.

2. NETWORK DESIGN

A star topology was built using one Cisco 2960 switch and six PCs. Each PC was connected via straight-through copper cables.

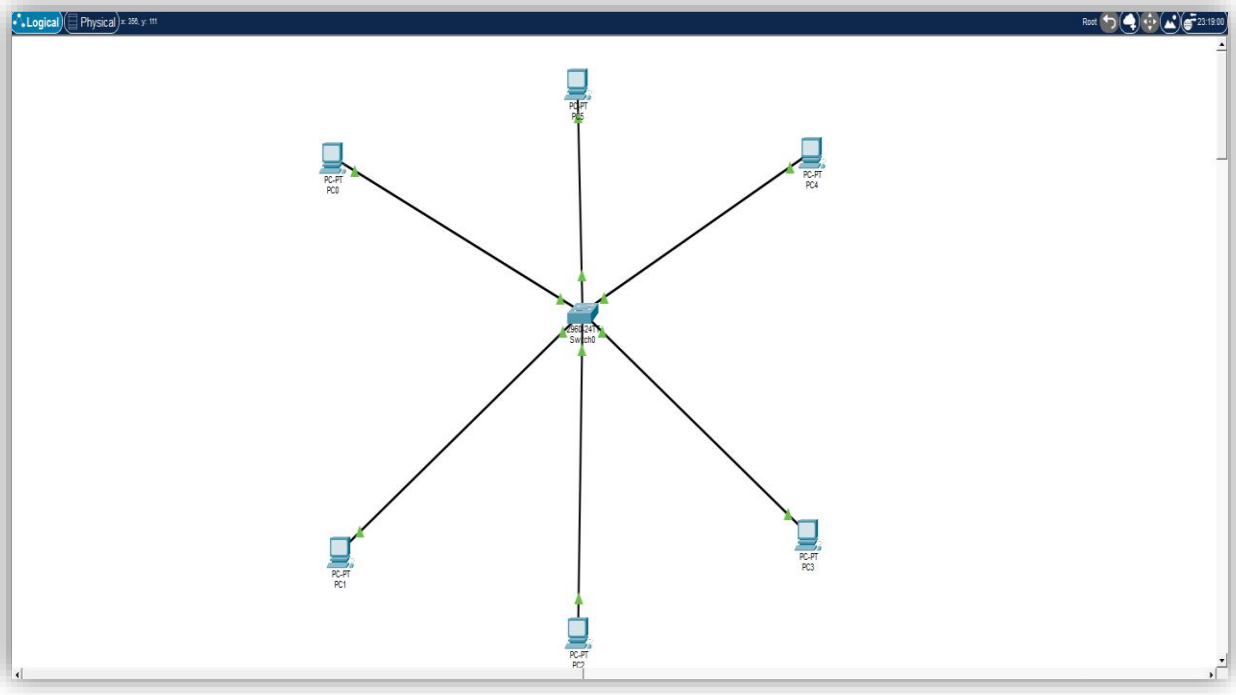


Figure 1: Network Setup (Star topology LAN)

3. IP CONFIGURATION

Each PC received a static IP from the `192.168.1.0/24` subnet as shown in Table 1.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
PC5	192.168.1.6	255.255.255.0

Configurations were applied via "Desktop → IP Configuration" on each PC. The switch operated with default settings.

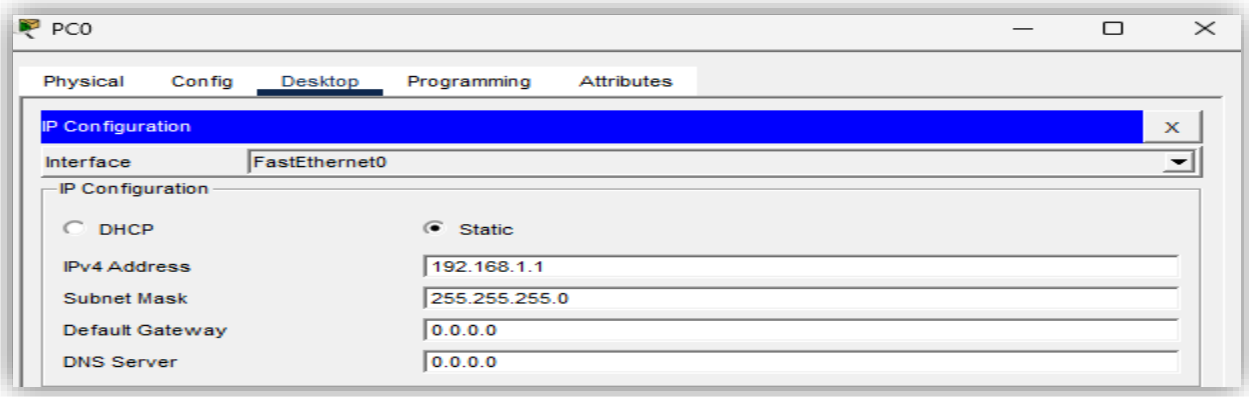


Figure 2: IP Configuration for PC0

4. CONNECTIVITY TEST

A `ping` from PC0 (192.168.1.1) to PC3 (192.168.1.4) was successful, confirming basic connectivity.

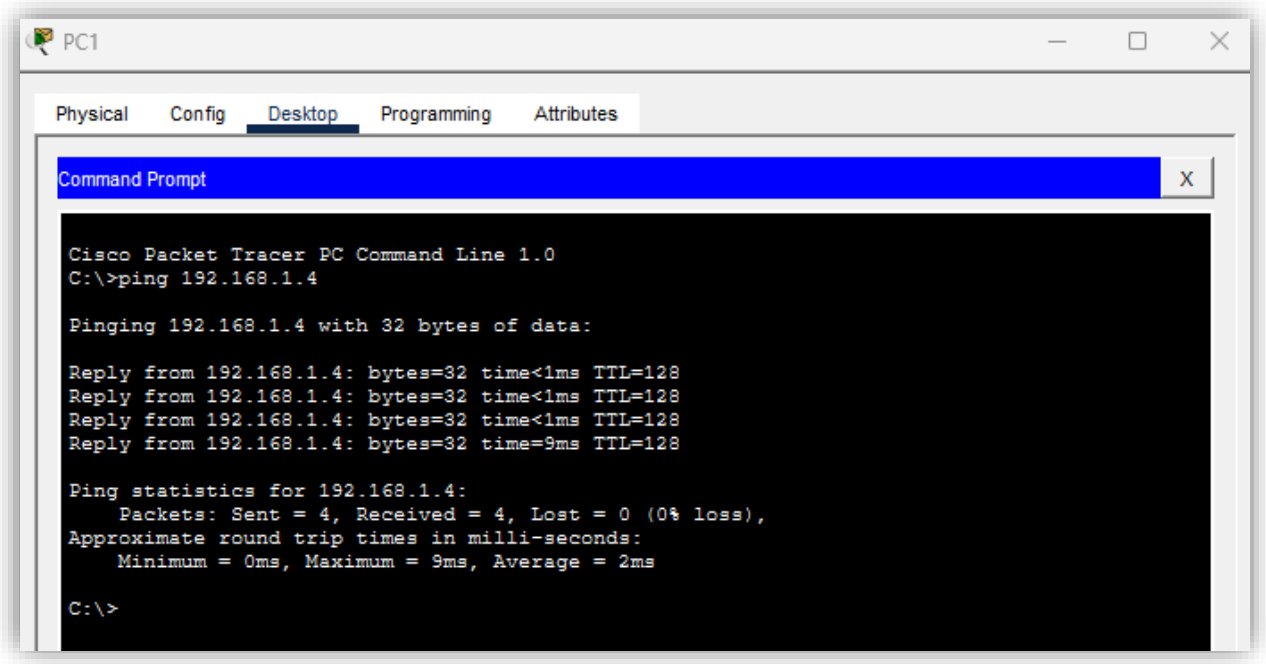


Figure 3: Connectivity test between PCs using ping

5. SIMULATION MODE – PACKET FLOW ANALYSIS

To fulfill the key objective of running the simulation, Packet Tracer's simulation mode was enabled. An ICMP packet from PC1 to PC3 was captured.

The observed packet flow:

1. PC1 generates an ICMP echo request.
2. The switch receives the frame and forwards it only to the port connected to PC3 (based on MAC address table).
3. PC3 replies with an ICMP echo reply, retracing the same path.

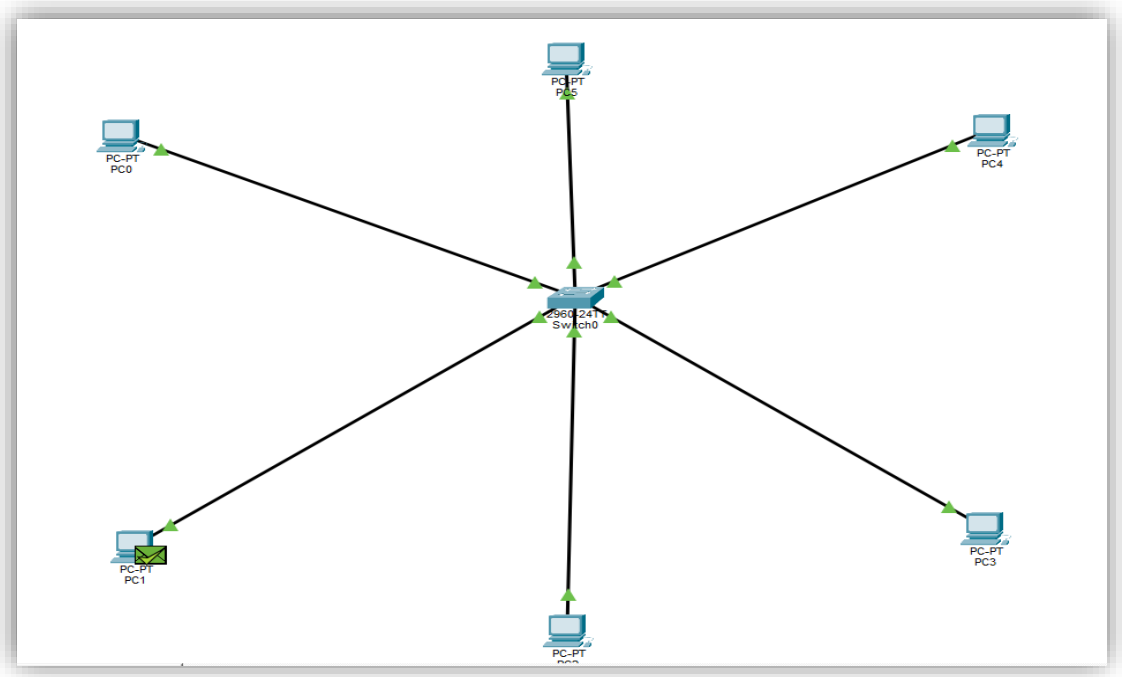


Figure 4: Successful stimulation testing between PC1 and PC3

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC1
	0.002	Switch0
	0.003	PC3
	0.004	Switch0
	1.999	--
Visible	2.000	Switch0
Visible	2.000	Switch0
Visible	2.000	Switch0
Visible	2.000	Switch0
Visible	2.000	Switch0
Visible	2.000	Switch0

Figure 5: Result in the Stimulation panel

This simulation visually confirmed that the switch intelligently forwards unicast traffic without flooding all ports.

6. KEY FINDINGS

- The star topology was established without any cabling or configuration errors.
- Full connectivity was achieved, enabling communication between any two PCs.
- Simulation mode revealed the stepwise movement of ARP and ICMP packets across the network.
- The switch demonstrated selective forwarding, transmitting frames only to the specific destination port.
- Ping responses were consistently fast (average latency between 0–2 ms) with no packet loss.

7. ANALYSIS

The experiment highlighted the operational advantages of a switched star topology. By dynamically building a MAC address table, the switch minimized unnecessary traffic and improved bandwidth utilization. Activating Simulation Mode in Packet Tracer offered a layer-by-layer visualization of data transmission, bridging theoretical OSI model concepts—particularly the switching function at Layer 2 and IP addressing at Layer 3—with practical implementation.

6. CONCLUSION

The star topology LAN was successfully implemented. IP configurations were applied, connectivity was verified via ping, and the simulation mode clearly demonstrated the end-to-end packet path through the switch. All lab objectives, including the simulation requirement, were fully met.